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Experiment – 7

K-Means Clustering for Customer Segmentation

Theory of K-Means Clustering

Concept of Clustering

Clustering is an unsupervised machine learning technique used to group a set of data points into clusters such that points within the same cluster are more similar to each other than to points in other clusters. Unlike supervised learning, clustering does not rely on labeled data — it discovers hidden patterns or natural groupings purely from the structure of the input data.

Clustering is widely applied in customer segmentation, anomaly detection, document categorization, image compression, and biological data analysis. The goal is to maximize intra-cluster similarity (compactness) while maximizing inter-cluster dissimilarity (separation).

Working of the K-Means Algorithm

K-Means is a partitioning-based clustering algorithm that divides n data points into k non-overlapping clusters. The algorithm follows an iterative process:

- Step 1 — Initialization: Choose k initial centroids. The k-means++ strategy selects centroids that are spread apart, which leads to better and more consistent convergence compared to random initialization.
- Step 2 — Assignment: Assign each data point to the nearest centroid based on Euclidean distance. Each point belongs to exactly one cluster.
- Step 3 — Update: Recalculate each centroid as the mean of all data points assigned to it.
- Step 4 — Repeat: Repeat the Assignment and Update steps until centroids no longer change significantly (convergence), or until the maximum number of iterations is reached.

The algorithm converges when cluster assignments no longer change between iterations. K-Means is sensitive to the initial placement of centroids and to the scale of features, which is why StandardScaler is applied before clustering.

Role of Centroids

A centroid is the geometric center (mean position) of all data points within a cluster. Centroids serve as the representative point of each cluster. During each iteration, centroids are recalculated as the mean of all assigned points, causing them to shift towards the true center of the cluster.

After convergence, the final centroids represent the average characteristics of each customer segment. For example, a centroid with high Annual Income and high Spending Score represents the high-income, high-spending customer group. Analyzing centroids in the original feature space provides direct business insights into each cluster.

Key Concepts

WCSS (Within-Cluster Sum of Squares)

WCSS, also called inertia, measures the total squared distance between each data point and the centroid of its assigned cluster. It is defined as:

$$WCSS = \Sigma \Sigma ||x_i - \mu_j||^2$$

Where x_i is a data point and μ_j is the centroid of its cluster. A lower WCSS indicates that the data points are tightly packed around their centroids (compact clusters). WCSS always decreases as k increases, which is why it is used alongside the Elbow Method to find the optimal k rather than simply minimizing it.

Elbow Method

The Elbow Method is a heuristic technique for selecting the optimal number of clusters k. It involves running K-Means for a range of k values (e.g., k = 1 to 10) and plotting the WCSS against k. As k increases, WCSS decreases — but the rate of decrease slows down.

The "elbow point" is where the rate of decrease sharply changes, forming a visible bend in the curve. This point indicates the optimal k, beyond which adding more clusters yields diminishing returns. In the Mall Customers dataset, the elbow typically appears at k = 5, suggesting 5 natural customer groups based on Annual Income and Spending Score.

Silhouette Score

The Silhouette Score is a quantitative metric that measures both the cohesion and separation of clusters. For each data point i, it is computed as:

$$s(i) = (b(i) - a(i)) / \max(a(i), b(i))$$

Where a(i) is the average intra-cluster distance (mean distance from point i to all other points in the same cluster), and b(i) is the average nearest-cluster distance (mean distance from point i to all points in the nearest neighboring cluster).

The Silhouette Score ranges from -1 to +1. A score close to +1 means the point is well-matched to its own cluster and poorly matched to neighboring clusters. A score near 0 indicates the point is on or near the decision boundary between two clusters. A negative score indicates possible misclassification.

The average Silhouette Score across all data points is used to compare different values of k. A higher average score indicates better-defined, well-separated clusters.